

FINM3403 GROUP ASSIGNMENT 1

Topic: Stock Market Reaction to the Announcement of an Equity Raising

1. Overview

This group assignment asks you to conduct an event study measuring how the stock market reacts to the *announcement* of a seasoned equity offering (SEO) — a sale of additional common shares by an already-listed company. You will replicate, on a smaller scale, the core approach of a classic result in Corporate Finance, apply it to a sample of real SEOs drawn from SDC Platinum, a database that tracks financial transactions, and matched to stock prices from the Center for Research in Security Prices (CRSP) database, and compare your findings to the published result.

By the end of this assignment, you should be able to:

- Explain why a SEO announcement might move a stock's price, and what economic theories predict the direction and magnitude of that move.
- Match a corporate event to the correct point in a firm's raw trading history and build an event window around it from scratch.
- Compute market-adjusted abnormal returns and cumulative abnormal returns (CARs) around a corporate event.
- Test whether the average CAR is statistically distinguishable from zero.
- Test the hypothesis that the size of an offering is related to the magnitude of the price reaction.
- Critically compare your own empirical results to a published study, and reason about *why* your results might differ (sample period, sample composition, index choice, methodological choices).

2. Motivation

Firms raise a large share of their capital from external sources. However, firms are reluctant to issue new common stock, despite equity issuance being a straightforward way to raise funds. Economists Paul Asquith and David Mullins (1986, *Journal of Financial Economics* see *lecture 3 folder for this reading*) investigate this issues and test whether the *announcement* of a SEO itself moves the issuing firm's stock price and investigate the reasons.

They suggest three possible reactions to the announcement:

- **No price effect.** If the firm's shares have close substitutes and markets are efficient, the number of shares a firm chooses to sell shouldn't affect its price. Only expected future cash flows should.
- **A negative price effect**, consistent with several distinct arguments: a downward-sloping demand curve for the firm's own shares (selling more of them requires a lower price to place them all with investors); changes to the firm's capital structure (less leverage can reduce the value of the associated tax shield); or an *information* story, where investors

infer that managers with private information are more likely to issue equity when they believe the stock is overvalued, so the announcement itself is bad news.

- **A positive price effect**, if investors instead interpret the raise as signalling a valuable investment opportunity, or that the reduction in leverage lowers the firm's expected costs of financial distress.

Asquith and Mullins test the above predictions on 531 registered common stock offerings by industrial and utility firms in the period 1963-1981 and find a clear, statistically significant *negative* announcement-day reaction; roughly -2.7% for industrial issuers and a smaller -0.9% for utilities. Furthermore, they show that, for industrial firms, the size of the reaction scales with the size of the offering. This assignment asks you to test whether a similar pattern holds in a sample of recent U.S. SEOs.

3. Market-Adjusted Returns

Asquith and Mullins measure abnormal performance using a single-index (market) model, which requires estimating each firm's own alpha and beta by OLS regression over a separate estimation window.

This assignment requires you team to use the simpler approach of *market-adjusted returns* instead.

The abnormal return on a SEO by firm i on day t is defined as $AR_{i,t} = R_{i,t} - R_{m,t}$ where $R_{i,t}$ is the daily return on the stock and $R_{m,t}$ daily return on the market index (the S&P 500).

This approach implicitly assumes every firm has a market beta of exactly 1 and an alpha of exactly 0. It is a strong simplification relative to the approach taken by Asquith and Mullins. It requires no regression, no estimation window, no per-firm parameter estimates. The cumulative abnormal return (CAR) over the event window is simply the sum of the daily abnormal returns (ARs) in that window.

4. The Data

This is a **group assignment**. Each team is assigned one of **three sample versions** of the data which comprises of **two files**. The group-sample assignments are listed below. For illustrative purposes, Group #10, will use sample 2 for their analysis. The data on Blackboard is posted to the relevant folders.

Groups Numbers	Sample Assigned	File Names
Groups 1-9	1	seo_sample_version_1_marketadj.csv seo_sample_version_1_marketadj_crsp_data.csv
Groups 10-18	2	seo_sample_version_2_marketadj.csv seo_sample_version_2_marketadj_crsp_data.csv
Groups 19-27	3	seo_sample_version_3_marketadj.csv seo_sample_version_3_marketadj_crsp_data.csv

Every version of the sample is built the same way from the same SDC Platinum / CRSP-matched pipeline, but each contains a different, entirely non-overlapping set of 500 SEO events. No event appears in more than one version.

Every event in every version of the sample

- Is a “follow-on” common stock offering (i.e., not an IPO, rights issue, or convertible security)
- Is a U.S. issuer listed on the NYSE, AMEX, or Nasdaq at the time of the offering (no OTC-listed firms)
- Has a clean, unambiguous match to a single CRSP “PERMNO”, the firm-identifier used in the database that we extract stock price and returns data from¹
- Has a complete 5-day announcement window of return data
- Has already been screened for duplicate/overlapping filings by the same firm, a minimum \$5 average share price in the run-up to the offering, and a small number of extreme statistical outliers

The *first* file, `seo_sample_version_N_marketadj.csv` (where **N** is the sample version, 1, 2 or 3), containing SEOs has 500 rows, one per event. The columns are

Column (Variable) Name	Definition
SEO_ID	Unique identifier for this equity offering
PERMNO	CRSP security identifier for the issuing firm
filing_dt	SDC filing date for the offering (i.e., event date)
Issuer_Borrower_Name_Full	Issuing firm's name
SDC_Deal_Number	SDC Platinum deal identifier
Gross_Proceeds	Dollar proceeds of the offering
utility_flag	1 if the issuing firm is a regulated utility, 0 otherwise
financial flag	1 if the issuing firm is a financial firm, 0 otherwise
real estate flag	1 if the issuing firm is a real estate firm, 0 otherwise
sample_version	sample version number of SEO data

This file tells you *what* happened and *when* — it does **not** contain any stock price data and does not contain any pre-computed results (no CAR, no market equity, no offering dilution). Computing all of that yourselves, from the raw price data below, is the point of Tasks 1–6 described later in the document.

The *second* file, `seo_sample_version_N_marketadj_crsp_data.csv`, contains daily stock price and return data and is structured differently to the first file. It is the *raw* daily trading history for every distinct firm in your assigned sample (i.e., first file), one row per firm per

¹ In analyses such as this, stock tickers are never used as firm identifiers as they are not unique and can be reused. Case in point is FB/META, a firm that has used different tickers. An analysis that uses tickers as the firm identifier will assume that they are two different firms.

trading day, spanning roughly two years before to two years after that firm's own SEO announcement date(s) in your sample (a firm with more than one SEO in your sample gets a single continuous span covering all of its own events, not a separate file per event). This file does not contain a pre-built event window, i.e., there is no column marking which row is “day 0” (i.e., the event day) “day -2” or any other position relative to an announcement — that alignment is something *you* need to build, using the “filing_dt” from the events file described above.

The columns in *second* file are described in the table below.

Column (Variable) Name	Definition
PERMNO	CRSP security identifier — join this to the events file
DLYCALDT	Calendar date of this observation
DLYRET	The stock's daily total return ($R_{i,t}$)
DLYPRC	The stock's daily closing price
SHROUT	Shares outstanding, in thousands (see note below)
SPRTRN	The S&P 500 daily return — the market benchmark ($R_{m,t}$)
VWRETD	CRSP value-weighted market daily return (provided for reference only; not used in this assignment)
TRADINGSTATUSFLG	Flags whether the stock was actively trading that day

A small number of firms will have less than the full two years of history on one side or both. This happens when a firm's stock didn't exist yet in CRSP before the start of the requested window (e.g. the firm is a recent IPO). This is a genuine data limitation, not an error in the file or on your part: if it affects one of your events, note it in your report and decide how to handle it (e.g. exclude that specific calculation for that event, or work with whatever history is actually available).

Shares outstanding note: “SHROUT” is reported by CRSP in *thousands* of shares, not actual share counts. Anywhere you compute a dollar market value of equity (price × shares outstanding, correctly scaled), keep this in mind — an unscaled calculation will be off by a factor of 1,000, which will distort the results of Task 4(b) and Task 6.

A note on returns: “SPRTRN” is a **price** return (it does not include dividends²) of S&P 500 index, while ‘DLYRET’ is a **total** return (it includes dividends). This is a known, minor mismatch — keep it in mind when interpreting your results, but it should not meaningfully affect your CARs over such short windows.

² Recall from introductory finance that the stock return has two components: capital gains and dividends. A price return only captures capital gains.

5. Identifying day 0 and Getting Data

Because the file containing stock prices and returns is not pre-aligned to any event, the first real work you need to do is build the event window, for every event in your sample.

- Take an event's "PERMNO" and "filing_dt" from the events file (i.e., first file).
- In the stock price file (i.e., first file), find all rows for that "PERMNO".
- Find the first 'DLYCALDT' on or after "filing_dt" — this is **day 0** for that event. If a firm has more than one event in your sample, do this separately for each event's own "filing_dt". The raw price rows are shared across that firm's events, but each event gets its own day 0.
- Counting outward from day 0 in actual **trading days present in the data** and not calendar days; weekends and holidays aren't trading days and shouldn't be counted as one. Assign relative-day positions: day 0, day -1 (the trading day immediately before it), day +1 (immediately after), and so on out to -10/+10 for the wide window and -2/+2 for the tight window (see Section 6).
- Do this for **every event** in your sample before moving on to Task 2.

This is genuine, event-study-relevant work. Asquith and Mullins, and every event study since, had to do exactly this kind of matching before any abnormal return could be computed.

Writing one clean, reusable piece of code that does this matching for all 500 events at once rather than handling events one at a time by hand and will save you a great deal of time in the tasks that follow.

6. Building Two Event Windows

This assignment uses **two different windows around the announcement**, and it matters which one you use where:

- **The wide window, (-10, +10):** 21 trading days, from 10 days before the filing date to 10 days after. Used only for the descriptive day-by-day picture in Task 3, it shows you the market's reaction over time, including any drift before or after the announcement itself.
- **The tight window, (-2, +2):** 5 trading days, centred on the announcement. **This is the window that defines your actual sample.** An event only belongs in your final analysis if it has a complete 5-day tight window of return data. This is also the window used for Task 4's announcement CAR, its significance test, and the offering dilution calculation.

In short: the wide window is for description; the tight window is for everything that counts as "the result."

7. **Task 1 of Assignment: Identify and Import the Data**

For every event in your sample, carry out the matching process described in section 5: locate the correct “PERMNO’s” stock price/returns history, find that event’s day 0, and confirm you can align the surrounding trading days around it. Report basic descriptive statistics on your matched sample: how many unique firms, how your events are distributed over time, and anything unusual you find including any events where a firm's available price history doesn't fully cover the window you need (see the note in Section 4), and how you have handled those issues.

8. **Task 2 of Assignment: Narrow the data to the event windows**

Using the *day 0* positions you built in Task 1, construct the two event windows described in Section 6: the wide (-10, +10) window and the tight (-2, +2) window. Confirm, for each event, that you have a complete tight window (5 observations). An event with any missing day inside (-2, +2), whether from a data gap or insufficient price history, should be treated as ineligible for the analysis. Report how many events in your sample have a complete tight window.

9. **Task 3 of Assignment: Abnormal and Cumulative Abnormal Returns (-10, +10)**

Compute the daily abnormal return, $AR_{i,t}$, for every event on every day of the wide window, using the formula in Section 3. Then produce a day-by-day summary table, one row per event day from -10 to +10, reporting:

- the mean AR across all events on that day
- the number of events (N) contributing to that day
- a t-statistic and p-value testing whether the mean AR is significantly different from zero
- the cumulative average abnormal return (CAR) up to and including that day (the running sum of the mean daily ARs)

This is what Asquith and Mullins report in their Table 2. Present your table and briefly describe its features: is there any drift before the announcement? On which day(s) does the reaction concentrate? Does the market keep drifting afterward, or stabilise?

10. **Task 4 of Assignment: Abnormal and Cumulative Abnormal Returns (-10, +10)**

Using the **tight (-2, +2) window only**, compute each event's announcement CAR (the sum of its 5 daily ARs). Then report and discuss the following

- (a) Report the mean and median announcement CAR across your sample and formally test whether the mean is statistically different from zero (state your null hypothesis, test statistic, and conclusion). Compare your result, in direction and rough magnitude, to Asquith and Mullins's own findings -2.7% industrial and -0.9% utilities.
- (b) Offering dilution is concerned with the amount of shareholder value that was destroyed by the equity offering. Asquith and Mullins capture the impact by estimating the announcement-day reaction in dollar terms, relative to the size of the deal.

For each event, compute the pre-announcement market value of equity (price × shares outstanding on day -11, the day just before your wide window starts and use it to express the dollar value of the price reaction as a percentage of the offering's gross proceeds³:

$$Dilution_i = \frac{CAR_{i,(-2,+2)} \times \text{Market Value of Equity}_{i,-11}}{\text{Gross Proceeds}_i}$$

Report the mean *and* median of offering dilution across your sample. (Hint: Think about what happens to this ratio when Gross Proceeds is very small relative to firm size.) For context, Asquith and Mullins report -31% (mean) and -28% (median) dilution for primary industrial issues.

11. **Task 5 of Assignment: Comparing Industrial vs. Utility Firms**

Asquith and Mullins separately analyse industrial and public utility issuers and find that utilities show a smaller announcement-day reaction, which they attribute to utilities being more regulated, more homogeneous, and their financing needs more predictable and already anticipated by investors.

Using the “utility_flag” column, split your sample into utility and non-utility (“industrial”) firms and compare their announcement CARs (from Task 4). Report the mean CAR and standard deviation for each group and formally test whether the two groups’ mean CARs differ (state your test, its assumptions, and your conclusion). Also compare the two groups’ variability — is one group more or less variable than the other?

Be upfront in your report about sample size here: utilities are a small share of the overall population this sample was drawn from, so your utility subgroup may be too small for a properly powered test of the mean difference on its own. Report the group sizes, describe what you find directionally (in both the mean and the variability comparison), and discuss what a larger utility subsample might be needed to say more confidently.

12. **Task 6 of Assignment: Is the size of the offer related to the market reaction?**

Asquith and Mullins's central cross-sectional finding is that, for industrial issuers, the announcement-day price reaction is significantly related to the *size* of the offering (measured as offering proceeds divided by pre-announcement firm equity value). Even in their own paper, this result is not uniform: their Table 7 finds a clearly negative, significant coefficient for “All” and “Primary” offerings, but a much weaker one for “Secondary” offerings and essentially none at all for “Combination” offerings. Test the same basic relationship in your sample

Construct a measure of relative offering size for each event, “offer size”, using the same market equity figure from Task 4(b). Offer size is defined as

$$\text{Offer size}_i = \frac{\text{Gross Proceeds}_i}{\text{Market Equity}_i}$$

2. Regress your announcement CAR estimated in Task 4a on this size measure:

³ Don't forget about the scaling issue with the number of shares outstanding (SHROUT) described in 4.

$$CAR_{i,(-2,+2)} = \alpha + \beta Offer Size_i$$

- Report your estimates of the OLS, i.e., *alpha*, *beta*, their significance, and the R^2 . Interpret the sign and magnitude of beta on its own terms — don't assume in advance which direction it should point to.

Note: this is a simplified, one-variable version of the regression in Asquith and Mullins's Table 7. You are not required to include their second regressor (the previous eleven-month cumulative excess return, "CER"). Leaving it out is a real simplification, not a cosmetic one — keep it in mind when comparing your coefficient to theirs in Task 7.

13. **Task 7 of Assignment:** The Discussion.

Put your team in the shoes of analysts at a prestigious investment bank. You are asked to write an explanatory note to your clients (i.e., CFOs) who are considering a round of capital raising. The aim of the report is to explain on how investors tend to react to announcements of seasoned equity issues by firms.

You leverage the analysis described in Tasks 3-6 to describe magnitude and direction of the market's reaction.

Your report should also include following:

- You can compare your findings to previous known results (i.e., Asquith and Mullins). If your numbers differ, what are some plausible reasons (different sample period, different market benchmark, different industry composition, the market-adjusted-vs-single-index methodological choice itself)? If your Task 6 coefficient comes out a different *sign* than Asquith and Mullins's, that is a real, interesting result worth discussing on its own terms, not a sign of an error — note that even their own Table 7 doesn't find a uniform result across every subgroup of their own data.
- Where your team's results are not statistically significant (this is expected for some case, especially in Task 5), what does or doesn't the result mean? Note that a non-significant result with sound methodology is not a failed result.
- Asquith and Mullins can't distinguish between the *price-pressure* hypothesis (a downward-sloping demand curve for the firm's own shares) and the *information* hypothesis (investors treat an equity issue as a negative signal about firm value) — their tests are consistent with either. Does anything in your own results shed light on which explanation seems more plausible, or do you agree that your data can't distinguish between them either?
- Any limitations of the market-adjusted return methodology (Section 3) itself that you think are worth flagging.

14. The Deliverables

Submit the following:

1. The written report described in Task 7 and should include tables and graphs. **Submit your report as a single PDF file**, named with your team number, for e.g. “Team_10.pdf”. Please format this document using 1-inch margins, 1.5-line spacing, and 12-point Times New Roman font.

The total number of pages should not exceed 7 including tables and graphs or your time-constrained clients might lose interest. It might require multiple drafts to be fit all that is required in this assignment.
2. You can use any language/software (Excel, SAS, R, Python, Stata). Your code should be fully commented. The submission of the code should be in **PDF format**, and the file name should include the team number “Team_10_python_code.pdf”. Should you use Excel, please submit the file and follow the same file naming convention.
3. Your AI usage log. See below.

AI Usage Policy

Given the widespread usage of AI tools, I would expect it to be used in this context as well for **coding help, debugging, or explaining statistical concepts**. While this is realistic professional practice, and using these tools well is itself a skill worth developing, every group is fully responsible for understanding, explaining, and defending every number and every line of code in their submission.

To support this:

- **Submit a log of your AI prompts and key responses** alongside your report (a simple chat export or screenshot log is fine). This should reflect your actual usage, not a curated summary — if you used AI extensively, the log should show that.
- Each group has been assigned one of three unique 500-event sample (see the data file naming for your group). Answers, code, or write-ups that closely match another group’s — despite each group having different underlying data — will be treated as a shared work for both groups involved.